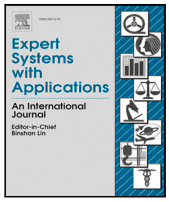
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Multi-population differential evolution approach for feature selection with mutual information ranking

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ABSTRACT

Feature selection is a crucial aspect of data preprocessing because of the significant effect of redundant features on classification performance and the extensive computational resources required. Evolutionary algorithm-based feature-selection methods have shown remarkable results in determining the optimal feature subset. To further enhance classification performance, this paper proposes a novel multi-population differential evolution approach for feature selection with mutual information ranking (MI-MPODE). Firstly, population preprocessing guided by mutual information is employed to reduce the dimensionality of the initial feature space. Then, the feature subset obtained from mutual information serves as the initial population for MI-MPODE. MI-MPODE incorporates a novel multi-population information-sharing mechanism, with common individuals from three layers contributing to inter-subpopulation information sharing. Additionally, an individual enhancement strategy is proposed to handle the variations of individuals in the population, and a Lens imaging opposition-based learning method is adopted to improve the algorithm's optimization capability. MI-MPODE is compared with several state-of-the-art evolutionary algorithm-based feature selection methods through experimental comparisons. The results show that MI-MPODE outperforms all comparison algorithms on more than half of the datasets, with a significant reduction in the number of features used, demonstrating a significant advantage over competitors.

1. Introduction

Classification problems have been widely investigated and are prevalent across various domains, such as product recommendation systems (Lin et al., 2022), medical diagnosis (Alweshah et al., 2022), spam and information detection (Jalal Mussa & M. Jameel, 2019), and remote sensing image classification (Zhang et al., 2018). However, rapid advancements in digital technology have generated significant amounts of high-dimensional data. Hence, data management has become exceptionally challenging when using classical classification algorithms, as it results in increased computational costs and storage requirements. In this regard, feature selection (FS) is key in solving classification problems. FS not only assists in reducing the dimensionality of the feature space but also enhances the accuracy and robustness of classification models. Additionally, it allows one to better understand the characteristics and structure of the data, as well as provides valuable insights for data analysis and model interpretation.

Based on evaluation metrics, FS methods can be categorized into filter, wrapper, and embedded methods. Filters primarily utilize relevant evaluation metrics to conduct statistical or correlation analyses of features to select the optimal feature subset. Typical filter methods include variance analysis, information theory, and correlation coefficients (Thabtah et al., 2020). In contrast to filter methods, wrapper-based methods embed FS into machine-learning classification algorithms such as K-nearest neighbors (KNN) (Sun et al., 2022), support vector machines (SVMs) (Tao et al., 2019), and decision trees (DT) (Liu & Zhao, 2012). Embedded methods combine FS with classifier training (Lu, 2019), thus allowing feature importance and model performance to be considered simultaneously. However, different methods may yield distinct classification performances for different datasets and problems. Therefore, integrating different FS methods can leverage the advantages of each method to further enhance classification performance.

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Classical FS methods present certain limitations as they typically feature strict requirements for feature-evaluation metrics or model selection. FS, which aims to select the fewest features while achieving good classification performance, can be considered an optimization problem. Hence, FS methods based on evolutionary computation (EC) have garnered widespread attention (Xue et al., 2016). Various EC techniques have been applied to FS, *e.g.*, particle swarm optimization (PSO) and differential evolution (DE), which use vector representations; ant colony optimization (ACO) with a graph structure (Chen et al., 2020); genetic programming (GP) with a tree structure (Nag & Pal, 2016); and sparse matrices with matrix representations (Li et al., 2022). Moreover, EC techniques have been combined with filter, wrapper, and hybrid methods to develop innovative FS approaches.

(1). ***EC-based filter methods:*** Filter methods typically rely on statistical or information-theoretical measures to gauge the optimal feature subset. Compared with other approaches, these evaluation methods are more intuitive and easier to understand. In the information-theory domain, Hancer et al. (2018) used mutual information (MI) to measure the correlation between features and class labels. They developed an MI-based FS method that focused on the maximum relevance and minimum redundancy. This method was integrated into the DE for FS. Similarly, Usman et al. (2020) utilized a binary cuckoo search algorithm and introduced two distinct multi-objective filter FS frameworks that utilized nondominated sorting genetic algorithms. In terms of distance measurement, Yang et al. (2012) performed neighborhood component analysis (NCA) to address high-dimensional FS problems. They verified that their algorithm was insensitive to irrelevant features. To further evaluate the NCA method, Hancer et al. (2023) incorporated NCA into the objective function of DE, thereby introducing an evolutionary filter FS method that is applicable to both single- and multi-objective scenarios. Meanwhile, Xue et al. (2019) enhanced the performance of PSO in FS by employing a population initialization based on the relief method. Regarding other filter methods, Li et al. (2009) proposed a multi-population agent genetic algorithm for feature selection to address the FS problem, where they used correlation measures to evaluate the relationship between features and labels. Chakraborty (2008) posited that the selection of an optimal feature subset depends on the objective function used for evaluating the feature subset. Hence, they introduced a feature-selection optimization algorithm based on PSO and the fuzzy set theory. Despite the good adaptability of EC-based filter methods, they still face challenges when dealing with high computational complexity data. Furthermore, this approach requires specific problem and data considerations, along with parameter setting, to achieve optimal effectiveness.

(2). ***EC-based wrapper methods:*** The most-investigated approach combines the global search capability of EC with classifiers to achieve the optimal feature subset. However, in single-objective optimization, no method is suitable for all scenarios. Hence, Feng et al. (Tan et al., 2008) proposed a GA-based feature subset-selection method that integrates multiple FS approaches. Zorarpacı and Özel (2016) combined the binary DE with the artificial bee colony algorithm to devise a new binary mutation strategy for DE. Additionally, they introduced an enhanced onlooker-bee process to address the FS problem. To enhance the optimization performance of algorithms in FS, Ouadfel and Abd Elaziz (2020) improved the local and global search capabilities during FS using an adaptive perceptual balance crow search algorithm. Qiu and Liu (2021) allocated PSO particles to three layers and employed a random-sample selection strategy and Gaussian-based elite strategy to maintain the exploration and exploitation of the feature search space. Owing to its dual objective of reducing feature count while enhancing classification accuracy, multi-objective optimization is more suitable for FS. Xue et al. (2014) first applied multi-objective DE to FS, where they regarded FS as a bi-objective optimization problem. Meanwhile, Nguyen et al. (2017) introduced a new FS decomposition strategy (based on the decomposition of multi-objective evolutionary algorithms), which improved classification performance. Agrawal et al. (2023) employed a multi-objective multimodal DE to select the optimal feature subsets. Consequently, Due to EC's global search capabilities, EC-based wrapper methods effectively enhance the performance of machine learning models. However, this method is computationally intensive, requiring multiple model trainings and significant dependency on algorithm parameters.

(3). ***EC-based hybrid methods:*** As filters are more effective in managing redundant features and wrappers exhibit better classification performance, most researchers have combined these two methods with EC to further enhance classification performance. For example, to address the challenges of hyperspectral image classification, Ghosh et al. (2013) developed a feature subset-selection method based on adaptive differential evolution. They employed a relief algorithm to rank the features and eliminate duplicates. The performance of the final selected feature subset was evaluated using the KNN algorithm. Determining the optimal number of features in a subset is challenging in cases involving high-dimensional data. Tran et al. (2019) proposed a variable-length representation method for FS based on PSO. Before employing PSO, they ranked the features based on distance metrics to achieve a smoother fitness landscape. The optimal feature subset for FS is not unique and selecting an appropriate number of features is crucial for achieving high classification accuracy. Therefore, Wang et al. (2023) used feature clustering to facilitate evolutionary FS. During feature clustering, they employed the maximum information coefficient to classify the features and evaluated the classification accuracy using the KNN algorithm. Compared to the previous two methods, EC-based hybrid methods exhibit significant advantages by integrating the strengths of other methods to adapt to various problems. However, while this method enhances performance, it also increases computational complexity. It requires a thoughtful integration of various methods and parameters to leverage its advantages effectively.

Based on the above issues, since DE is a structurally simple and powerful optimization algorithm, its mutation and crossover operations help maintain good diversity within the population, thereby enhancing feature diversity in feature selection. Meanwhile, to further investigate multi-population mechanisms and expand the research of DE in feature selection. Therefore, this paper proposes a novel algorithm named MI-MPODE, which incorporates a multi-population information-sharing mechanism. First, this paper uses MI techniques to preprocess the feature space before executing MI-MPODE, thus eliminating redundant and weakly correlated features to enhance the efficiency and interpretability of the selected feature subsets. Second, the information-sharing mechanism among the subpopulations allows the algorithm to escape from local optima during optimization. To further improve the performance, this paper categorizes individuals in each subpopulation into three layers using an individual enhancement strategy to enhance optimization. Additionally, this paper integrates a lens imaging opposition-based learning (LensOBL) strategy to strengthen the diversity of the feature subsets.

The main contributions of this study are as follows:

- A novel multi-population information-sharing mechanism is proposed to reduce the probability of the algorithm falling into local optima.
- The utilization of MI technology to optimize the initial population of the algorithm effectively reduces the initial feature space.

- The proposal of a three-layer individual enhancement strategy to improve solution accuracy. Additionally, a LensOBL strategy is employed to enhance the diversity of the feature subsets.
- The performance of MI-MPODE is evaluated comprehensively by testing it on multiple FS datasets. The evaluation assesses not only its effectiveness in solving FS problems but also the performance of the newly introduced strategies.

The remainder of this paper is organized as follows: Section 2 presents the relevant background information, including the DE framework, opposition-based learning (OBL) strategy, and MI. The proposed MI-MPODE method is described in detail in Section 3. Section 4 discusses the experimental results and analyses of different algorithms for solving the FS problem, where the effectiveness of the newly introduced strategies is confirmed via the ablation experiment. Finally, Section 5 provides the conclusions and some perspectives pertaining to the current study.

2. Background

2.1. DE

DE is a simple yet effective evolutionary algorithm introduced by Storn and Price (Storn & Price, 1997); furthermore, it is well known for its simplicity and minimal parameter requirements. DE primarily comprises initialization, mutation, crossover, and selection operations. The detailed structure of the associated algorithm is as follows.

(1). **Initialization:** In classic DE, a population comprises individuals randomly generated from continuous values between the maximum and minimum boundaries. Herein, *rand* represents a random number between $(0, 1)$.

$$X_i^0 = L_b + rand \cdot (U_b - L_b) \tag{1}$$

where L_b and U_b are the minimum and maximum boundary values of the initial search space, respectively, and X_i^0 represents the ith individual in the initial population NP, which is the total number of individuals in the population.

(2). **Mutation:** For the target vector V_i^t, three vectors distinct from the target vector are randomly selected within the population, thus resulting in the generation of a mutant vector V_i^{t+1}.

$$V_i^{t+1} = X_{r1}^t + F \cdot (X_{r2}^t - X_{r3}^t) \tag{2}$$

where $r1$, $r2$, and $r3$ are distinct random integers within the interval [1,NP]; and $F \in (0,1)$ is a mutation factor. The mutation strategy "$DE/rand/1$" is used in Eq. (2).

(3). **Crossover:** For each dimension of the individual, the crossover operator CR is used to decide whether to select the target vector $X_{i,j}^t$ or mutant vector $V_{i,j}^{t+1}$, thus ultimately forming a trial vector $U_{i,j}^{t+1}$.

$$U_{i,j}^{t+1} = \begin{cases} V_{i,j}^{t+1}, & \text{if } rand(0,1) \le \text{CR} \parallel j = randi(1, D) \\ X_{i,j}^t, & \text{otherwise} \end{cases} \tag{3}$$

where $rand(0, 1)$ is a random number between (0,1); and $randi(1, D)$ is a random integer within the interval [1,D], with D representing the population dimension.

(4). **Selection:** Based on the objective function, the fitness of the trial vector U_i^{t+1} and target vector X_i^t is evaluated, and a new generation of individuals X_i^{t+1} is obtained via greedy selection.

$$X_i^{t+1} = \begin{cases} U_i^{t+1}, & \text{if } f(U_i^{t+1}) \le f(X_i^t) \\ X_i^t, & \text{otherwise} \end{cases} \tag{4}$$

where $f(*)$ is the fitness value of the current vector. During the minimization process, if the fitness value of the trial vector is less than that of the target vector, then the target vector is replaced; otherwise, the target vector is retained.

2.2. MI

MI (Shannon, 2001), which is a concept from Information theory, is used to quantify the relationship between two random variables and indicates the amount of information that one variable can provide regarding the other. MI can be regarded as the shared uncertainty between two random variables. If the mutual information $I(X;Y)$ between two vectors X and Y is 0, then one can infer that these two variables do not provide any information regarding each other. MI can be applied in many fields, such as FS, pattern recognition, and image registration. In the context of FS, a higher MI value indicates a stronger correlation between the feature vector and class label.

(1). **Entropy (H):** Entropy measures the uncertainty of a random variable. Therefore, for a discrete variable X, its entropy $H(X)$ is defined as.

$$H(X) = -\sum_{x \in X} P(x) log P(x) \tag{5}$$

where $P(x)$ is the probability of variable X. In other words, entropy can be regarded as the average amount of information for a random variable. An uncertain event with a greater probability contains more information. In a two-dimensional space, the MI value between two random variables is obtained from their joint and conditional mutual information, which can be defined as.

$$H(X,Y) = -\sum_{x \in X, y \in Y} P(x,y) log P(x,y) \tag{6}$$

$$H(X|Y) = -\sum_{x \in X, y \in Y} P(x,y) log P(x|y) \tag{7}$$

(2). **MI:** MI measures the dependence or correlation between two random variables. The MI for two discrete random variables X and Y can be defined as.

$$I(X;Y) = \sum_{x \in X} \sum_{y \in Y} P(x,y) log\left(\frac{P(x,y)}{P(x)P(y)}\right) \tag{8}$$

when MI is combined with entropy, it can be represented as the entropies of X and Y. Consequently, Eqs. (6) and (7) can be written as $I(X;Y) = H(X) + H(Y) - H(X,Y)$ and $I(X;Y) = H(X) - H(X|Y)$, respectively, when expressed in terms of entropy.

2.3. OBL

OBL, which is introduced by Tizhoosh (2005), is an effective optimization strategy that aims to select both the target point and its opposite point. Integrating OBL with EC allows the fitness of both the target vector and its opposite vector to be evaluated simultaneously. This integration not only enhances the population diversity of the algorithm but also accelerates its convergence speed. Moreover, owing to the increase in studies pertaining to OBL over the past two decades, many improved methods related to OBL have been proposed and the theoretical effectiveness of OBL has been demonstrated (Mahdavi et al., 2018). The classic OBL is expressed as follows:

$$x'_{i,j} = a_{i,j} + b_{i,j} - x_{i,j} \tag{9}$$

where $x_{i,j}$ is the jth dimension of the ith target vector and $x'_{i,j}$ is its opposite vector; meanwhile, $a_{i,j}$ and $b_{i,j}$ are the maximum and minimum values of the jth dimension for all vectors, respectively. In EC, the vector closest to the optimal solution is retained based on the fitness of the target vector and its opposite vector.

3. Proposed algorithm

3.1. Population initialization based on MI

This study evaluated the MI values between each feature and class label. A higher MI value indicates a strong correlation between the

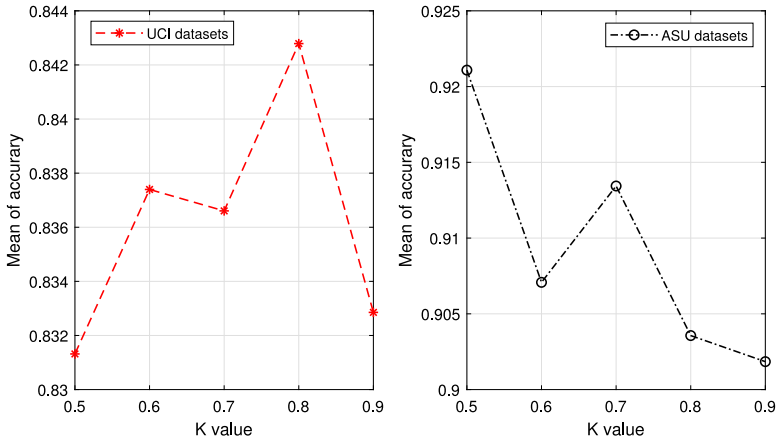


Fig. 1. The impact curve of different K values on average classification accuracy across multiple datasets.

features and class labels. This paper used these MI values to rank the features and select a subset of features with higher MI values. Subsequently, the selected feature subset was used as the initial feature space for MI-MPODE. This method not only initially reduces the dimensionality of the feature space but also helps improve classification performance and reduce computational costs. The main role of it can be verified in the experiments of Section 4.5. The pseudocode for population initialization based on MI is presented in Algorithm 1.

Algorithm 1 Population initialization based on MI

Require: Datasets, Feature vector (X), Label vector (y), Threshold (K);
Ensure: Optimal feature subset;
1: **for** $i = 1$ to Num_Feat **do**
2: Calculate the MI value for each feature using Eq. (8);
3: **end for**
4: Sort the MI values of all features in descending order;
5: Select the top K features as the optimal feature subset and use them as the initial population;

Setting the parameter K to represent the number of selected features is crucial in this approach. Different datasets may require different numbers of features, thus resulting in different final results. Therefore, the appropriate balance must be determined. To ensure that MI maximally filters out redundant features in different datasets without affecting the final classification accuracy, our research suggests setting the filtering ratio K in the range of $[0.5, 0.9]$. The main reasons are as follows: if K<0.5, a large number of features would be selected, among which there is a possibility of non-redundant features, thereby interfering with the final classification accuracy. On the other hand, when K>0.9, too few features are selected, leaving many redundant features, which limits the effectiveness of MI. Therefore, this paper tested the average classification accuracy of five different K values on the 20 datasets presented in Section 4.1 (see Fig. 1). As shown in Fig. 1, the best average classification accuracies were achieved using $K = 0.8$ and 0.5 for small-scale UCI and large-scale ASU datasets, respectively. Therefore, this paper selected K values of 0.8 and 0.5 for small- and large-scale data, respectively.

3.2. Multi-population information sharing mechanism

Multi-population mechanisms are effective for solving global optimization problems (Wang et al., 2024). Each subpopulation represents an independent search space and has its own optimal solution; additionally, it compensates for the disadvantage of single populations, i.e., ease of being trapped in local optima. Efficient information sharing among subpopulations is crucial for achieving a global optimal solution within multiple populations. For example, in earlier studies, the differential evolution based on a multi-population mutation strategy ensemble (MPEDE) achieved significant results (Wu et al., 2016). Inspired by MPEDE, Sun et al. (2020) proposed a multi-population based creating ideas ensemble and embedded the basic framework of the brain storm optimization algorithm within it, enhancing the algorithm's search capability. To further improve the performance of MPEDE, Li et al. (2021) proposed a multi-population cooperation and multi-strategy integration DE (MPMSDE). However, unlike previous studies, MPMSDE divides the original population into several unequal subpopulations and employs a grouping and ranking strategy to allocate different mutation strategies to the subpopulations, thereby reasonably allocating computational resources while increasing population diversity. Additionally, multi-population approaches are also commonly used in multi-objective optimization. Zhao et al. (2019) divided the population of the non-dominated sorting genetic algorithm 2 (NSGA-II) into multiple subpopulations and assigned different crossover operators to each subpopulation, enhancing the algorithm's adaptability to different problems. As current scheduling methods for workflow scheduling in heterogeneous distributed computing environments still need improvement, Xie et al. (2023) adopted a two-stage multi-population genetic algorithm to address this issue, thereby improving the search efficiency of the solution space.

To further investigate the information sharing methods among multiple populations, this paper analyzed several population-based optimization algorithms, such as chicken swarm optimization (CSO) (Meng et al., 2014), grey wolf optimization (GWO) (Mirjalili et al., 2014), artificial fish swarm algorithm (AFSA) (Neshat et al., 2014), and bacterial foraging optimization (BFO) (Simon, 2008). Research showed that these algorithms comprise multiple subpopulations and individuals of different categories within the entire population. These individuals assume distinct roles in performing their respective operations. For example, the foraging process of three types of chicken swarms in CSO, the hunting process of four types of wolf swarms in GWO, and the flow and clustering operations in BFO. They seek optimal solutions through collaboration and information exchange among individuals and different subpopulations, thus providing certain advantages for solving various types of optimization problems.

Based on the aforementioned description, this paper categorizes individuals in the DE population into three levels: elite, common, and inferior. The core of the information-sharing mechanism lies in common individuals. These individuals not only learn from elite individuals in different subpopulations but also from common individuals in other subpopulations, thereby achieving information sharing. Specifically, when an common individual updates, it learns from elite individuals and other common individuals in different subpopulations. If a subpopulation becomes trapped in a local optimum, during the next subpopulation update, it learns beneficial genes from other subpopulations, thus escaping the local optimum. Hence, this paper constructed

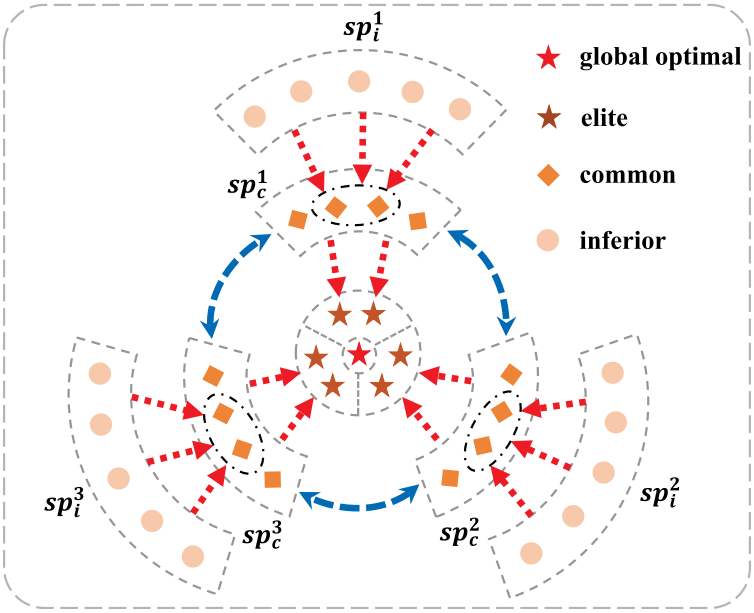


Fig. 2. Model of multi-population and information sharing mechanism.

a novel multi-population hierarchical DE (MI-MPODE) using the model shown in Fig. 2. In Fig. 2, the depth of the icon color represents the fitness of the subpopulations. The common individual (rhombus) learns from the elite individual (pentagon) located at the center, whereas the blue bidirectional arrows represent information sharing among common individuals from different subpopulations. The inferior individual (circle) learns only from its mother individual (the individual inside the elliptical dashed line) while maintaining the population's update. In Fig. 2, three subpopulations are assumed to exist in the population, where sp_i^j and sp_c^j represent the inferior and common individuals of the jth subpopulation, respectively, with $j \in [1, 2, 3]$. Specifically, the number of subpopulations in this paper is 3, and detailed explanations can be found in Section 4.2.

3.3. Layered individual update and enhancement

Each subpopulation contained three types of individuals: elite, common, and inferior. Each type had a unique update strategy. In addition, to ensure both the exploration and exploitation of the population, this paper adopted different enhancement strategies for different individuals to improve the performance of the algorithm. According to relevant studies (Cai et al., 2018), common individuals constitute the largest proportion of society, whereas the elite and inferior individuals constitute only a small fraction. Meanwhile, the latest research sets the ratio of the three types of individuals as 2:5:3 (Guan et al., 2024), which was successfully validated for performance improvement. However, due to the smaller population size in this study, setting common individuals at 50% would result in no significant distinction among the three types of individuals. Therefore, this paper set the ratio of the elite, common, and inferior individuals to 2:6:2. The update methods for each type of individual are described in detail below.

Elite individuals are vital to global optimization algorithms as they determine whether the algorithm can obtain a global optimal solution. To maintain the superior genes of elite individuals, this paper applied a slight Gaussian perturbation to the elite individuals and determined their fitness to generate different perturbations. Furthermore, no other operations were performed on them. This method largely preserves the superior genes of elite individuals, enabling the algorithm to rapidly approach the global optimum and accelerate convergence speed. The update method for the elite individuals is expressed as follows:

$$X_i^{t+1} = X_e^t \cdot (1 + N(\mu, \sigma)) \tag{10}$$

$$\sigma = \begin{cases} 1, & \text{if } f(X_i) \le f(X_j) \\ exp(\frac{f(X_j) - f(X_i)}{f(X_i) + \tau}), & \text{otherwise} \end{cases} \tag{11}$$

where X_e^t is an elite individual, and $N(\mu, \sigma)$ is a Gaussian distribution with $\mu = 0$, where σ is determined by the corresponding judgment condition; $f(*)$ is the fitness value of the current individual; τ is an infinitesimally small value to prevent the denominator from being 0.

In this paper, information sharing between subpopulations occurs primarily among the common individuals. As described in Section 3.2, the common individuals can learn from the elite and common individuals from other subpopulations. Therefore, the update of common individuals is composed of three components, i.e., the self-vector, elite learning vector, and common learning vector, where different learning weights are set for the learning vectors. The advantage of this method lies in that when a subpopulation falls into a local optimum, it does not cause the entire population to fall into a local optimum. Each subpopulation ensures the overall excellence of the population through continuous learning of common individuals. In addition, because the update method of common individuals is similar to the mutation operation of DE, no additional mutation operations are performed, only crossover operations are carried out. The update method for common individuals is expressed as follows:

$$V_i^{t+1} = X_i^t + F_i^t \cdot (X_e^t - X_i^t) + (1 - F_i^t) \cdot (X_c^t - X_i^t) \tag{12}$$

where X_c^t is a common individual; and F_i^t and $(1 - F_i^t)$ are learning factors for acquiring knowledge from elite individuals X_e^t and other common individuals X_c^t, respectively.

During the iteration process of the algorithm, the population loses diversity, which causes the algorithm to fall into local optima. To maintain the diversity of the population while the inferior individual learns from its "mother" common individual, logic mapping is performed to generate a larger search range for the inferior individual. In addition, similar to common individuals, this process is similar to the mutation operation of DE. Therefore, and no additional mutation operations are performed, only crossover operations. Thus, the update method for the inferior individuals is expressed as follows:

$$\begin{aligned} V_i^{t+1} &= X_i^t + L_i^t \cdot (X_m^t - X_i^t) \\ L_i^{t+1} &= \varphi \cdot L_i^t \cdot (1 - L_i^t) \end{aligned} \tag{13}$$

where X_m^t is the common individual who is the mother of the current inferior individual; and L_i^t is a learning factor generated via logistic mapping, whose value is a random number in the interval (0,1). Compared with a random distribution, logistic mapping can result in more chaotic phenomena, thus increasing the search range of inferior individuals.

Moreover, to enhance the performance of the algorithm, this paper applies an adaptive strategy (Qin, 2006) to update both common and inferior individuals. Specifically, the learning factor (F_i^t) for common individuals is adaptive and generated by Eq. (14), while the learning factor (L_i^t) for inferior individuals is generated by logical mapping. Similarly, both undergo crossover operations, and the crossover operator is adaptively generated by Eq. (15). This method allows the offspring to inherit good genes from their parent individuals, thus further improving the performance of the algorithm. The adaptive strategy is expressed as follows:

$$F_i^{t+1} = \begin{cases} F_l + rand \cdot F_u, & \text{if } rand \le \phi \\ F_i^t, & \text{otherwise} \end{cases} \tag{14}$$

$$CR_i^{t+1} = \begin{cases} rand, & \text{if } rand \le \psi \\ CR_i^t, & \text{otherwise} \end{cases} \tag{15}$$

where F_i^{t+1} and CR_i^{t+1} represent the learning factor (i.e., mutation operator) and crossover operator for the ith individual in the $(t+1)$ generation, respectively; $rand$ is a random number in the interval (0,1); F_l and F_u are the minimum and maximum values of the learning factor, which are set to 0.1 and 0.9, respectively; and both ϕ and ψ are set to 0.1.

3.4. Lens imaging opposition-based learning

In our previous research (Yu et al., 2024), lens imaging opposition-based learning (LensOBL) has been validated as an effective optimization strategy. Compared with OBL, LensOBL is more flexible in addressing optimization problems. OBL itself searches for opposite points in symmetric spaces, whereas LensOBL avoids the issue of losing better solutions due to symmetry by introducing parameters r and k, thus offering the population more opportunities to obtain the optimal solution. The model for the population-selection strategy based on LensOBL is illustrated in Fig. 3, and its mathematical expression is as follows. From Fig. 3, it is easy to derive Eq. (16). Eq. (17) is the main implementation formula for the opposite solution of LensOBL, which can be obtained by transforming Eq. (16). For the detailed transformation process, please refer to the original literature.

$$k = \frac{h}{h'} = \frac{\frac{a+b}{2} - x}{x' - \frac{a+b}{2}} = \frac{r}{x' - \frac{a+b}{2} - r} \tag{16}$$

$$x' = (1 + \frac{1}{k}) \cdot r + \frac{a+b}{2} \tag{17}$$

where a and b are the boundary values of the one-dimensional search space in Fig. 3. In the original literature, h and h' are respectively referred to as the height of the candle and the height of the image presented. It is noteworthy that r is the search radius, which can be determined by the target point x_i and the current boundary of the search space $[a, b]$. k, as a scaling factor, is a variable that dynamically changes according to the interval of the target point x_i. Further details can be found in Yu et al. (2024). All variable values are consistent with those in the original literature. Moreover, in OBL, the opposite point x_i' is not necessarily better than the target point x_i. Therefore, to avoid this phenomenon, researchers set a jump probability Jr, and the OBL strategy was executed within the probability range, which reduced the computational cost of the algorithm and increased the convergence rate. Similarly, LensOBL also adopts it, and the setting of Jr in this paper is consistent with the original literature.

Algorithm 2 FS method based on MI-MPODE

Require: Datasets, initial mutation operator (F), initial crossover operator (CR), Population size (NP), Dimension (D), Maximum number of iterations (Max_Iter);
Ensure: Optimal feature subset;
1: Obtain the initial population and evaluate its fitness according to Algorithm 1;
2: **while** iter $\le Max_Iter$ **do**
3: Sort the population and divide it into subpopulations;
4: **for** sp_i to subpopulations **do**
5: Update the elite, common, and inferior individuals respectively
using Eq. (10), Eq. (12) and Eq. (13);
6: Perform crossover operation through Eq. (3);
7: Adopt the LensOBL strategy according to Eq. (17);
8: Simultaneously evaluate the original and opposite individuals of
the subpopulation, and select offspring individuals by Eq. (4);
9: **end for**
10: iter = iter + 1;
11: Generate mutation and crossover operators for the next generation
using Eq. (14) and Eq. (15);
12: Obtain the feature subset and evaluate it using a classifier;
13: **end while**
14: Acquire the optimal feature subset and classification accuracy;

3.5. FS method based on MI-MPODE

In summary, this paper proposed a multi-population DE algorithm for FS that combines MI and LensOBL. The pseudocode for MI-MPODE to address the FS problem is detailed in Algorithm 2. This paper initially removed redundant features using the MI technique and obtained the initial population of the algorithm, as shown in line 1 of Algorithm 2. Subsequently, this paper entered the main loop of MI-MPODE (lines 2–13). This paper sorted the entire initial population by fitness and partitioned the corresponding subpopulations. Subsequently, this paper iterated through each subpopulation and updated its individuals (lines 4–9). Next, as shown in line 5, this paper updated the FS method based on MI-MPODE via the three types of individuals within each subpopulation; this process is similar to the mutation operation. Subsequently, as shown in line 6, all individuals except the elite individuals were subjected to crossover. At this point, all individuals in the subpopulation adopted a probabilistic LensOBL strategy to enhance the quality of the subpopulation. Finally, this paper simultaneously evaluated the original and opposite subpopulations and greedily selected the offspring. To better inherit the genes from the parent individuals, this paper updated the mutation and crossover operators for the next generation (see line 11). At the end of line 12, the algorithm was completed, thus resulting in the final feature subset (optimal solution), which was evaluated using a classifier. After completing the iterations, this paper obtained the optimal feature subset and classification accuracy. To intuitively understand the application process of MI-MPODE to the FS problem, a flowchart is presented in Fig. 4, which corresponds to Algorithm 2.

4. Experimental results and analysis

4.1. Datasets

In this study, the performance of the algorithm was evaluated using 11 UCI datasets (Kelly et al., 2023) and nine high-dimensional ASU datasets (Li et al., 2018), both of which are typically used for FS. The

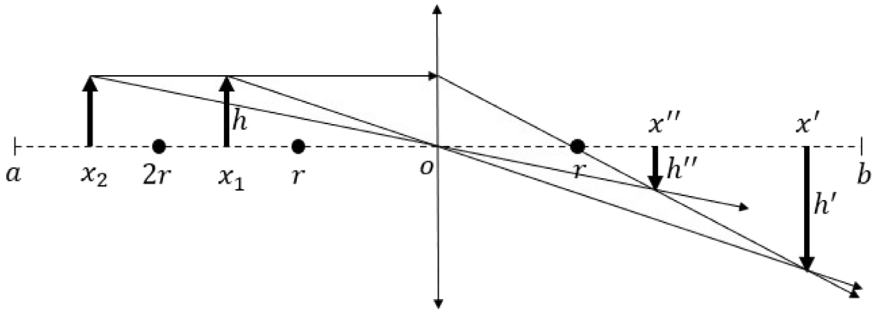


Fig. 3. The population selection strategy based on LensOBL.

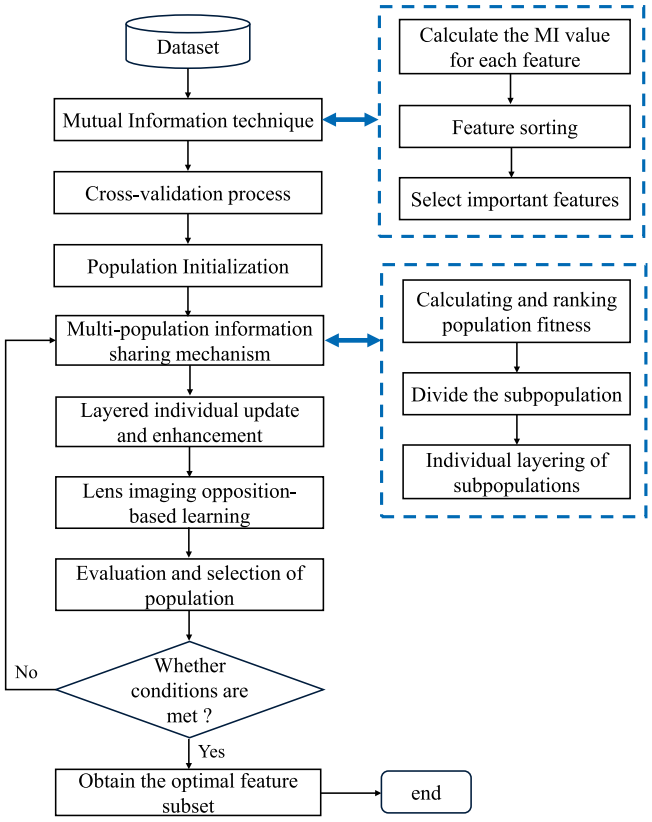


Fig. 4. Flowchart of MI-MPODE for solving FS problems.

relevant information for the datasets is provided in Table 1. Additionally, this paper defines UCI datasets with fewer than 500 features as small-scale datasets, while ASU datasets are referred to as large-scale datasets.

4.2. Experimental settings

Experiments were conducted in two stages. In the first stage, to ensure an unbiased experiment, this paper selected several state-of-the-art FS methods based on the EC. These methods included the classical PSO, DE, MSPSO (Qiu, 2019), COPSO (Too et al., 2022), TLPSO (Qiu & Liu, 2021), BASO (Too & Rahim Abdullah, 2020), ECSA (Ouadfel & Abd Elaziz, 2020), ISSA (Tubishat et al., 2020) and PLTVACIW-PSO (Azar et al., 2023). The second stage evaluated the effectiveness of the proposed method through ablation experiments. For all datasets in the experiments, this paper used a 10-fold cross-validation method to partition them into training and testing sets, with 70% and 30% of the samples selected for training and testing, respectively. To reduce

Table 1
The brief information of dataset.

Type	Name	Instances	Features	Classes
UCI datasets	Heart	303	13	5
	Zoo	101	16	7
	Parkinsons	195	22	2
	Dermatology	366	34	6
	Ionosphere	351	34	2
	LungCancer	32	56	3
	Movement-libras	360	91	15
	Musk1	476	166	2
	Arrhythmia	452	279	16
	LSVT	126	309	2
	SCADI	70	205	7
ASU datasets	Madelon	2600	500	2
	Yale	165	1024	15
	Colon	62	2000	2
	TOX-171	171	5748	4
	Prostate_GE	102	5966	2
	Leukemia	72	7070	2
	ALLAML	72	7129	2
	arcene	200	10 000	2
	GLI_85	85	22 283	2

algorithmic variability, each algorithm was executed 30 times independently, and each cross-validation process was mutually independent. Therefore, all experimental results were derived from the average of 30 independently executed trials. Furthermore, typical parameter settings were used for each EC algorithm, such as a population size (NP) of 10 and a maximum number of iterations (Max_Iter) of 100. Note that due to the small value of NP, and to reflect better information sharing among multiple populations, this paper sets the number of subpopulations to 3. This is mainly because too few subpopulations cannot effectively demonstrate the multi-population mechanism, while too many subpopulations would lead to too few individuals per subpopulation, thus hindering information sharing. Details regarding the other parameters are listed in Table 2. Finally, this paper used the KNN classifier to evaluate the classification accuracy of each method, with the number of neighbors (k) set to 5. The objective function for evaluating the algorithm is as follows:

$$Fitness = \alpha \cdot accuracy + (1-\alpha) \cdot \frac{N_s}{N_f} \tag{18}$$

where $accuracy$, N_s, and N_f represent classification accuracy, the number of selected features, and the total number of features, respectively; α and $(1-\alpha)$ are their respective weights.

The experimental environment for this study included Windows 11 with 128 GB of RAM and a Core(TM) i9-12900K CPU. The experimental code was developed using the MATLAB 2022b software.

4.3. Experimental results and analysis

This paper compared the classification accuracy, number of selected features, computation time, and achieved optimal fitness value of MI-MPODE with those of state-of-the-art algorithms in solving the FS

Table 2
The other parameter settings for state-of-the-art algorithms.

Algorithms	Year	Other parameters
PSO	–	$c_1 = c_2 = 2,\ w = 0.4.$
DE	–	$F = 0.5,\ CR = 0.9.$
MSPSO	2019	$c_1 = c_2 = 1.49445,\ w = 0.729.$
BASO	2020	$\alpha = 50,\ \beta = 0.2.$
ECSA	2020	$AP = 0.1,\ FL = 1.5.$
ISSA	2020	$c_1 = 2 * exp(-(4t/Max_Iter)^2),\ c_2 = c_3 = rand(0,1).$
PLTVACIW-PSO	2023	$c_{1,ini} = c_{2,end} = 1.05,\ c_{1,end} = c_{2,ini} = 1.28,\ w_{max} = 0.9,\ w_{min} = 0.4.$
TLPSO	2021	$c_1 = c_2 = 2,\ w = 0.9,\ \sigma = 0.3 - 0.2 * (t/Max_Iter).$
COPSO	2022	$c_1 = c_2 = 2,\ w_1 = 0.5 + 0.5 * rand,\ w_2 = 0.8 * rand.$
MI-MPODE	–	$Jr = 0.3,\ N_E : N_O : N_I = 2 : 6 : 2,\ \alpha = 0.95.$

Table 3
Comparative results of classification accuracy on datasets between MI-MPODE and state-of-the-art algorithms.

Name	Type	PSO	DE	MSPSO	COPSO	TLPSO	BASO	ECSA	ISSA	PLTVACIW-PSO	MI-MPODE
Heart	Mean	0.6157	0.6277	**0.6341**	0.6371	0.6097	0.6311	0.6082	0.4839	0.6157	0.6262
	Std	0.0276	0.0220	**0.0220**	0.0167	0.0417	0.0211	0.0323	0.0515	0.0253	0.0238
Zoo	Mean	0.9567	**0.9756**	0.9722	0.9689	0.9511	0.9600	0.9644	0.8333	0.9556	0.9700
	Std	0.0352	**0.0213**	0.0278	0.0230	0.0379	0.0282	0.0289	0.0510	0.0432	0.0282
Parkinsons	Mean	0.9075	0.9184	0.9224	0.9241	0.9034	0.9138	0.9172	0.8402	0.9190	**0.9224**
	Std	0.0262	0.0264	0.0263	0.0220	0.0316	0.0283	0.0312	0.0470	0.0253	**0.0221**
Dermatology	Mean	0.9860	**0.9956**	0.9903	0.9931	0.9879	0.9922	0.9888	0.8355	0.9854	0.9854
	Std	0.0112	**0.0068**	0.0103	0.0069	0.0102	0.0074	0.0075	0.1005	0.0100	0.0094
Ionosphere	Mean	0.9187	0.9067	**0.9406**	0.9054	0.9149	0.9330	0.8781	0.8375	0.9137	0.9346
	Std	0.0201	0.0181	**0.0206**	0.0215	0.0236	0.0207	0.0246	0.0289	0.0241	0.0188
LungCancer	Mean	0.9600	0.9600	0.9933	0.9867	0.9600	0.9867	0.8933	0.4533	0.9133	**0.9933**
	Std	0.0814	0.0814	0.0365	0.0507	0.0968	0.0507	0.1363	0.2097	0.1252	**0.0365**
Movement-libras	Mean	**0.8028**	0.7830	0.7907	0.7735	0.7787	0.7731	0.7531	0.7077	0.7901	0.8022
	Std	**0.0338**	0.0344	0.0344	0.0379	0.0372	0.0272	0.0315	0.0389	0.0403	0.0386
Musk1	Mean	0.9315	**0.9366**	0.9209	0.9096	0.9286	0.9134	0.9054	0.8338	0.9338	0.9305
	Std	0.0173	**0.0173**	0.0170	0.0230	0.0226	0.0220	0.0206	0.0366	0.0193	0.0200
Arrhythmia	Mean	0.6869	0.6778	0.6896	0.6627	0.6681	0.6743	0.6496	0.6020	0.6778	**0.6938**
	Std	0.0210	0.0139	0.0194	0.0143	0.0177	0.0157	0.0221	0.0268	0.0207	**0.0208**
LSVT	Mean	0.7090	0.6784	**0.8378**	0.7027	0.7072	0.8009	0.6820	0.6252	0.6766	0.8027
	Std	0.0588	0.0533	**0.0471**	0.0536	0.0568	0.0504	0.0723	0.0520	0.0629	0.0508
SCADI	Mean	0.7738	0.7833	0.8238	0.7857	0.7738	0.8071	0.7381	0.6929	0.7452	0.8333
	Std	0.0565	0.0439	0.0554	0.0622	0.0533	0.0626	0.0710	0.0799	0.0447	0.0342
Madelon	Mean	0.8012	0.7912	0.8174	0.7751	0.7961	0.7961	0.7626	0.6750	0.8109	**0.8442**
	Std	0.0173	0.0109	0.0203	0.0089	0.0155	0.0119	0.0116	0.0409	0.0194	**0.0171**
Yale	Mean	0.6694	0.6639	0.6660	0.6510	0.6585	0.6503	0.6272	0.5558	0.6701	**0.6966**
	Std	0.0417	0.0551	0.0413	0.0504	0.0645	0.0569	0.0587	0.0793	0.0531	**0.0519**
Colon	Mean	0.8528	0.9028	0.9722	0.8972	0.8500	0.9500	0.8778	0.8028	0.8556	**0.9806**
	Std	0.0715	0.0792	0.0456	0.1109	0.1012	0.0469	0.0922	0.1168	0.0874	**0.0420**
TOX-171	Mean	0.8033	0.8065	0.8216	0.7817	0.8007	0.7974	0.7680	0.7242	0.8072	**0.8601**
	Std	0.0410	0.0436	0.0498	0.0488	0.0361	0.0503	0.0481	0.0651	0.0545	**0.0471**
Prostate_GE	Mean	0.8967	0.8983	0.9533	0.9067	0.9000	0.9400	0.8933	0.8817	0.8867	**0.9600**
	Std	0.0601	0.0676	0.0414	0.0504	0.0572	0.0515	0.0716	0.0793	0.0765	**0.0443**
Leukemia	Mean	0.9476	0.9619	0.9952	0.9476	0.9429	0.9762	0.9286	0.9024	0.9333	**0.9952**
	Std	0.0591	0.0522	0.0181	0.0417	0.0544	0.0390	0.0622	0.0762	0.0725	**0.0181**
ALLAML	Mean	0.8857	0.8714	**0.9762**	0.8881	0.8810	0.9119	0.8881	0.8048	0.8690	0.9643
	Std	0.0764	0.0687	**0.0390**	0.0694	0.0781	0.0668	0.0719	0.0918	0.0624	0.0450
arcene	Mean	0.8717	0.8742	0.9342	0.8817	0.8817	0.9017	0.8767	0.8592	0.8750	**0.9308**
	Std	0.0419	0.0471	0.0368	0.0415	0.0580	0.0537	0.0572	0.0611	0.0525	**0.0352**
GLI_85	Mean	0.9275	0.9294	0.9863	0.9275	0.9098	0.9490	0.9157	0.9137	0.9373	**0.9885**
	Std	0.0550	0.0644	0.0253	0.0612	0.0506	0.0482	0.0668	0.0572	0.0511	**0.0239**

problem. Their detailed results are presented in Tables 3, 4, 5, and 6. In Tables 3 and 6, "Mean" and "Std" represent the mean and standard deviation of each algorithm's performance over 30 independent trials, respectively. The best solutions obtained by each algorithm on the dataset are indicated in bold.

As shown by the data in Table 3, MI-MPODE exhibited excellent classification accuracy across multiple datasets. Only PSO, DE, and MSPSO outperformed MI-MPODE on a few datasets. In the UCI datasets, although MI-MPODE achieved better classification accuracy than other algorithms on the "Parkinsons", "LungCancer", and "Arrythmia" categories, the difference compared with other algorithms was minimal. Notably, MSPSO, COPSO, TLPSO, and PLTVACIW-PSO showed significant improvements in terms of classification accuracy compared with PSO on several datasets, thus indicating their successful enhancements for solving FS problems (*e.g.*, "Heart", "Zoo", and "Ionosphere"). By contrast, the ISSA did not exhibit good classification performance compared with the other algorithms. In high-dimensional datasets, MI-MPODE did not achieve the best classification accuracy on "ALLAML" and "arcene", whereas MSPSO performed better on those datasets, respectively. Because MSPSO and MI-MPODE employ a multi-population mechanism and demonstrate good performance in solving high-dimensional FS problems, the multi-population mechanism can effectively address the problem of algorithms becoming trapped in local optima. In general, MI-MPODE showed significant improvements in terms of classification accuracy on multiple datasets. This is because the MI filtered out redundant features in the early stages of the algorithm, thus providing sufficient search space for the initial population; additionally, the improvement was due to the LensOBL maintaining the population diversity throughout the algorithm.

In FS, not only the classification accuracy but also the number of selected features is crucial. Based on the data in Table 4, MI-MPODE not

Table 4
Comparative results of the number of selected features between MI-MPODE and state-of-the-art algorithms on datasets.

Name	PSO	DE	MSPSO	COPSO	TLPSO	BASO	ECSA	ISSA	PLTVACIW-PSO	MI-MPODE
Heart	4.60	5.80	**3.33**	4.60	4.60	4.33	7.50	6.90	4.50	3.50
Zoo	6.43	7.37	5.87	7.03	6.80	7.53	9.90	8.10	6.70	**6.23**
Parkinsons	4.73	5.63	2.90	7.10	5.07	3.63	11.43	10.37	4.63	**2.83**
Dermatology	14.87	17.73	13.63	17.60	15.03	18.17	23.47	17.07	14.07	**11.67**
Ionosphere	11.20	16.77	**4.63**	11.83	12.57	4.87	19.53	17.13	12.13	4.67
LungCancer	14.67	22.97	4.60	20.70	16.23	8.30	33.43	27.43	15.10	**6.17**
Movement-libras	38.13	54.33	**21.10**	40.97	39.20	20.00	62.13	44.20	38.27	24.50
Musk1	76.47	111.30	46.47	83.53	77.87	56.57	132.90	83.43	75.23	**56.20**
Arrhythmia	124.97	190.93	**51.13**	138.27	124.07	50.03	202.07	129.23	112.87	65.60
LSVT	101.93	165.20	12.17	144.90	113.40	35.87	167.87	134.33	108.07	**6.37**
SCADI	64.67	97.63	18.03	89.53	68.17	28.97	112.37	99.27	61.47	**12.57**
Madelon	243.37	359.57	84.73	253.83	244.20	124.80	384.60	235.70	243.50	**84.53**
Yale	470.47	671.80	165.97	517.23	482.50	225.57	743.03	394.87	442.93	**160.70**
Colon	884.93	1035.23	43.33	938.80	893.10	249.93	1060.53	325.93	852.73	**25.67**
TOX-171	2776.20	3943.40	1139.10	2941.17	2822.30	2066.00	4534.40	1761.60	2737.43	**878.23**
Prostate_GE	2788.67	3041.97	91.87	2776.90	2815.37	814.60	3329.53	701.67	2735.93	**69.07**
Leukemia	3309.03	3756.17	87.10	3256.33	3351.97	643.57	3824.67	637.67	3258.73	**75.97**
ALLAML	3319.90	3698.50	160.10	3259.63	3359.03	778.20	3872.70	496.33	3268.53	**105.00**
arcene	4762.33	5693.80	737.47	4932.07	4808.10	1610.37	5766.27	1198.77	4719.63	**455.97**
GLI_85	10 628.93	12 146.27	850.90	10 576.03	10 824.30	4301.87	13 218.10	2115.33	10 697.67	**485.53**

only achieved the classification accuracy shown in Table 3 but also significantly reduced the feature dimensions. Furthermore, compared with other algorithms, it selected fewer features. Combining the data from Tables 3 and 4, this paper observed that some algorithms achieved were superiority in terms of either the classification accuracy or the number of selected features, but not in terms of both. For example, in the "Zoo" dataset, DE achieved the best classification accuracy, whereas MI-MPODE had the fewest selected features. In the "Movement-libras" dataset, PSO achieved the best classification accuracy, whereas MSPSO had the fewest selected features. In the "ALLAML" dataset, MSPSO and MI-MPODE each excelled in terms of the classification accuracy and the number of selected features, respectively. Therefore, MI-MPODE can achieve higher classification accuracies with fewer selected features. This further confirms that the MI can assist the algorithm in filtering redundant features, thus reducing the initial feature space and conserving computational resources.

Tables 5 and 6 present the computation times and fitness values of the best solutions for each algorithm, respectively. As shown in Table 5, MI-MPODE indicated relatively longer computation times, primarily because of the inclusion of the multi-population mechanism and LensOBL strategy, which increased the algorithm's computation time. By contrast, the original DE algorithm required a significantly lower computation time. Notably, MI-MPODE required the shortest computation time among the multiple high-dimensional datasets, thus confirming its suitability for solving high-dimensional FS problems. Because each algorithm's evaluation function is determined by a weighted combination of the classification accuracy and the number of selected features, some algorithms may exhibit differences in terms of the classification accuracy, number of selected features, and fitness values on the same dataset. Regarding the fitness values, the optimal solutions were primarily distributed between MSPSO and MI-MPODE on most datasets. The convergence curves for all algorithms on the selected datasets are shown in Fig. 5. Clearly, MI-MPODE consistently achieved good performance and showed descending trends on several datasets (*e.g.*, "SCADI", "Madelon", "Yale", and "TOX-171"). By contrast, several other algorithms such as PSO, DE, BASO, ECSA, and others were trapped in local optima, particularly on high-dimensional datasets (*e.g.*, "Colon", "GLI_85", and "Prostate_GE").

In summary, MI-MPODE achieved significant classification performance improvements compared with state-of-the-art algorithms across multiple datasets, particularly on high-dimensional datasets, where MI-MPODE outperformed the others. This can be attributed to the techniques introduced, including initial FS using MI to reduce redundancy and the LensOBL strategy to maintain the population diversity. However, MI-MPODE did not demonstrate its advantages on small-scale datasets. There are two main reasons for this: firstly, due to the fewer number of features in small-scale datasets, after the initial feature selection by MI technique and subsequent algorithmic filtering of redundant features, the remaining number of features becomes too few, thereby affecting the final classification accuracy. Secondly, the multi-population information sharing mechanisms and LensOBL strategy proposed in this paper are beneficial for maintaining population diversity. Therefore, when dealing with small-scale datasets, algorithms continuously maintain population diversity, thereby preventing rapid convergence to the global optimum solution.

4.4. Statistical test results

Nonparametric tests do not require assumptions about the distribution of data; instead, they are based on the order or rank of the data, which makes them more flexible. Therefore, to evaluate the performance differences of each algorithm in FS, this paper adopts two nonparametric test methods to assess the differences between algorithms, namely the Wilcoxon test and the Friedman test. Their results are presented in Tables 7 and 8, respectively. The wilcoxon test is typically used to compare differences between two paired samples, while the friedman test compares differences among multiple related samples. In Table 7, "R^+" and "R^-" respectively indicate whether MI-MPODE is superior to or inferior to the sum of rankings of the comparison algorithms. This paper selects a p-value of 0.05 as the significance level. Specifically, if the p-value is less than 0.05, it indicates a significant difference between MI-MPODE and the comparison algorithms, denoted as "Yes" for significance (sig.); otherwise, there is no significant difference. In Table 8, "*AVR*" and "*Rank*" respectively represent the ranking values obtained through the friedman test and the actual ranking values for easier observation of results. That is, the smaller the values of "*AVR*" and "*Rank*", the better the overall performance of the algorithm. For detailed information on non-parametric tests, please refer to the literature (Derrac et al., 2011).

According to the data in Table 7, it is evident that MI-MPODE exhibits no significant difference from MSPSO, and in Table 8, MI-MPODE ranks second only to MSPSO in overall performance. Through research, it is find that MSPSO also possesses a mechanism of multiple populations, thus enabling it to achieve good performance on the majority of datasets when compared to MI-MPODE. However, it is noteworthy that among all algorithms in Table 6, MSPSO's superior performance is concentrated on smaller datasets, while MI-MPODE demonstrates better performance on larger datasets. This also proves that multi-population

Table 5
Comparative results of the computation time of MI-MPODE and state-of-the-art algorithms on datasets.

Name	PSO	DE	MSPSO	COPSO	TLPSO	BASO	ECSA	ISSA	PLTVACIW-PSO	MI-MPODE
Heart	2.06	2.06	2.02	2.50	2.08	2.05	**2.01**	2.27	2.06	7.92
Zoo	1.95	1.94	1.97	2.35	1.96	**1.90**	1.88	2.18	1.96	9.42
Parkinsons	1.99	1.96	1.97	2.34	2.01	**1.93**	1.89	2.16	1.98	8.12
Dermatology	1.99	**1.98**	2.04	2.39	2.02	2.04	2.00	2.23	2.00	9.33
Ionosphere	2.01	**1.98**	2.10	2.40	2.01	2.01	**1.98**	2.18	2.00	7.71
LungCancer	1.84	1.82	1.87	2.21	1.87	1.85	**1.82**	2.05	1.86	3.04
Movement-libras	**2.07**	**2.07**	**2.07**	2.51	2.13	2.10	2.11	2.28	2.08	3.79
Musk1	**2.27**	2.30	2.19	2.74	2.35	2.32	2.40	2.44	**2.27**	3.97
Arrhythmia	2.50	2.57	**2.36**	3.01	2.62	2.53	2.70	2.55	2.46	4.01
LSVT	2.11	**2.01**	2.08	2.54	2.30	2.15	2.03	2.18	2.10	2.96
SCADI	1.99	**1.93**	1.98	2.43	2.11	2.02	1.94	2.12	1.99	3.04
Madelon	22.24	32.47	14.36	28.33	22.62	25.00	39.10	22.48	22.22	**13.91**
Yale	2.76	**2.44**	2.64	3.34	3.32	2.88	2.77	2.45	2.74	3.81
Colon	3.00	**2.17**	2.93	3.67	4.14	3.22	2.26	2.20	3.01	3.13
TOX-171	8.87	8.19	7.13	10.93	12.06	10.49	9.79	4.99	8.92	**5.36**
Prostate_GE	6.85	4.80	5.40	8.35	9.99	7.60	5.96	2.73	6.82	**3.95**
Leukemia	6.89	4.24	5.71	8.37	10.66	7.63	5.11	2.53	6.86	**3.68**
ALLAML	6.89	4.30	5.74	8.40	10.65	7.71	5.21	2.51	6.85	**3.62**
arcene	15.07	13.27	10.54	18.47	20.35	16.76	16.59	5.24	10.80	**6.97**
GLI_85	19.24	11.97	15.45	23.43	31.16	22.75	15.02	**4.66**	20.28	14.09

Table 6
Comparative results of fitness values between MI-MPODE and state-of-the-art algorithms on datasets.

Name	Type	PSO	DE	MSPSO	COPSO	TLPSO	BASO	ECSA	ISSA	PLTVACIW-PSO	MI-MPODE
Heart	Mean	0.3827	0.3760	**0.3604**	0.3625	0.3884	0.3671	0.4010	0.3774	0.3824	0.3726
	Std	0.0258	0.0216	**0.0204**	0.0158	0.0393	0.0207	0.0301	0.0243	0.0236	0.0209
Zoo	Mean	0.0613	0.0462	**0.0447**	0.0515	0.0677	0.0615	0.0647	0.0677	0.0632	0.0525
	Std	0.0312	0.0206	**0.0258**	0.0215	0.0344	0.0269	0.0274	0.0295	0.0391	0.0259
Parkinsons	Mean	0.0987	0.0903	**0.0803**	0.0882	0.1032	0.0902	0.1046	0.0930	0.0875	0.0816
	Std	0.0247	0.0240	**0.0248**	0.0206	0.0297	0.0260	0.0277	0.0246	0.0238	0.0208
Dermatology	Mean	0.0352	0.0302	**0.0292**	0.0324	0.0336	0.0341	0.0452	0.0427	0.0346	0.0355
	Std	0.0103	0.0069	**0.0091**	0.0069	0.0102	0.0096	0.0077	0.0107	0.0097	0.0098
Ionosphere	Mean	0.0937	0.1133	**0.0632**	0.1073	0.0993	0.0708	0.1445	0.0922	0.0999	0.0708
	Std	0.0207	0.0176	**0.0192**	0.0198	0.0239	0.0194	0.0217	0.0166	0.0233	0.0185
LungCancer	Mean	0.0511	0.0585	**0.0104**	0.0311	0.0525	0.0201	0.1312	0.0789	0.0958	0.0132
	Std	0.0775	0.0766	**0.0343**	0.0481	0.0916	0.0483	0.1276	0.0869	0.1192	0.0341
Movement-libras	Mean	0.2085	0.2363	0.2105	0.2380	0.2320	0.2266	0.2691	0.2414	0.2206	**0.2050**
	Std	0.0329	0.0316	0.0314	0.0351	0.0348	0.0264	0.0293	0.0370	0.0394	**0.0368**
Musk1	Mean	0.0881	0.0937	0.0891	0.1110	0.0912	0.0993	0.1299	0.1170	0.0855	**0.0871**
	Std	0.0174	0.0174	0.0158	0.0218	0.0220	0.0174	0.0174	0.0244	0.0186	**0.0199**
Arrhythmia	Mean	0.3198	0.3403	**0.3040**	0.3452	0.3375	0.3184	0.3691	0.3433	0.3263	0.3056
	Std	0.0206	0.0135	**0.0187**	0.0139	0.0173	0.0144	0.0197	0.0141	0.0196	0.0213
LSVT	Mean	0.2929	0.3322	**0.1560**	0.3058	0.2964	0.1949	0.3292	0.2081	0.3247	0.1895
	Std	0.0562	0.0503	**0.0446**	0.0509	0.0535	0.0486	0.0672	0.0530	0.0592	0.0480
SCADI	Mean	0.2307	0.2296	0.1718	0.2254	0.2315	0.1903	0.2762	0.2088	0.2570	**0.1644**
	Std	0.0532	0.0411	0.0511	0.0587	0.0511	0.0571	0.0661	0.0414	0.0420	**0.0308**
Madelon	Mean	0.2132	0.2343	0.1819	0.2390	0.2181	0.2062	0.2640	0.2450	0.2040	**0.1649**
	Std	0.0168	0.0106	0.0215	0.0086	0.0150	0.0112	0.0110	0.0211	0.0184	**0.0206**
Yale	Mean	0.3371	0.3521	0.3254	0.3568	0.3480	0.3432	0.3904	0.3606	0.3351	**0.3039**
	Std	0.0394	0.0527	0.0386	0.0478	0.0614	0.0520	0.0560	0.0557	0.0503	**0.0504**
Colon	Mean	0.1620	0.1182	0.0275	0.1211	0.1648	0.0537	0.1426	0.0554	0.1585	**0.0198**
	Std	0.0678	0.0745	0.0434	0.1058	0.0961	0.0444	0.0883	0.0471	0.0830	**0.0396**
TOX-171	Mean	0.2110	0.2181	0.1794	0.2330	0.2139	0.2105	0.2599	0.2060	0.2070	**0.1482**
	Std	0.0389	0.0412	0.0475	0.0464	0.0342	0.0428	0.0448	0.0433	0.0517	**0.0450**
Prostate_GE	Mean	0.1215	0.1221	0.0451	0.1119	0.1186	0.0638	0.1292	0.0531	0.1306	**0.0392**
	Std	0.0572	0.0645	0.0394	0.0475	0.0543	0.0488	0.0690	0.0476	0.0728	**0.0421**
Leukemia	Mean	0.0732	0.0628	**0.0051**	0.0728	0.0780	0.0272	0.0949	0.0255	0.0864	0.0056
	Std	0.0562	0.0496	**0.0173**	0.0395	0.0517	0.0370	0.0588	0.0402	0.0689	0.0171
ALLAML	Mean	0.1319	0.1481	**0.0237**	0.1292	0.1367	0.0891	0.1335	0.0967	0.1473	0.0354
	Std	0.0725	0.0647	**0.0369**	0.0657	0.0742	0.0628	0.0661	0.0581	0.0593	0.0424
arcene	Mean	0.1457	0.1480	**0.0662**	0.1371	0.1365	0.1015	0.1460	0.0835	0.1423	0.0703
	Std	0.0398	0.0438	**0.0355**	0.0392	0.0551	0.0506	0.0546	0.0351	0.0499	0.0337
GLI_85	Mean	0.0928	0.0943	0.0149	0.0927	0.1100	0.0581	0.1098	0.0285	0.0836	**0.0134**
	Std	0.0523	0.0609	0.0240	0.0580	0.0481	0.0480	0.0651	0.0419	0.0485	**0.0234**

approaches have significant advantages in improving algorithm performance. Furthermore, to facilitate the observation of data distribution and differences among several algorithms, box plots of several algorithms on some datasets are presented in Fig. 6. It can be clearly observed that MI-MPODE not only achieves optimal performance in these datasets but also has a median value smaller than other algorithms in multiple datasets (*e.g.*, "Movement_libras", "Madelon", "Yale", "Colon", and "Tox-171", etc.).

4.5. Ablation experiment

To validate the effectiveness of the proposed method, this paper primarily conducts ablation experiments on multiple population mechanisms, LensOBL, and MI techniques. Since the effectiveness and parameter sensitivity of LensOBL have been successfully validated in our previous research (Yu et al., 2024), this study does not conduct additional experiments solely on LensOBL. Furthermore, due to the

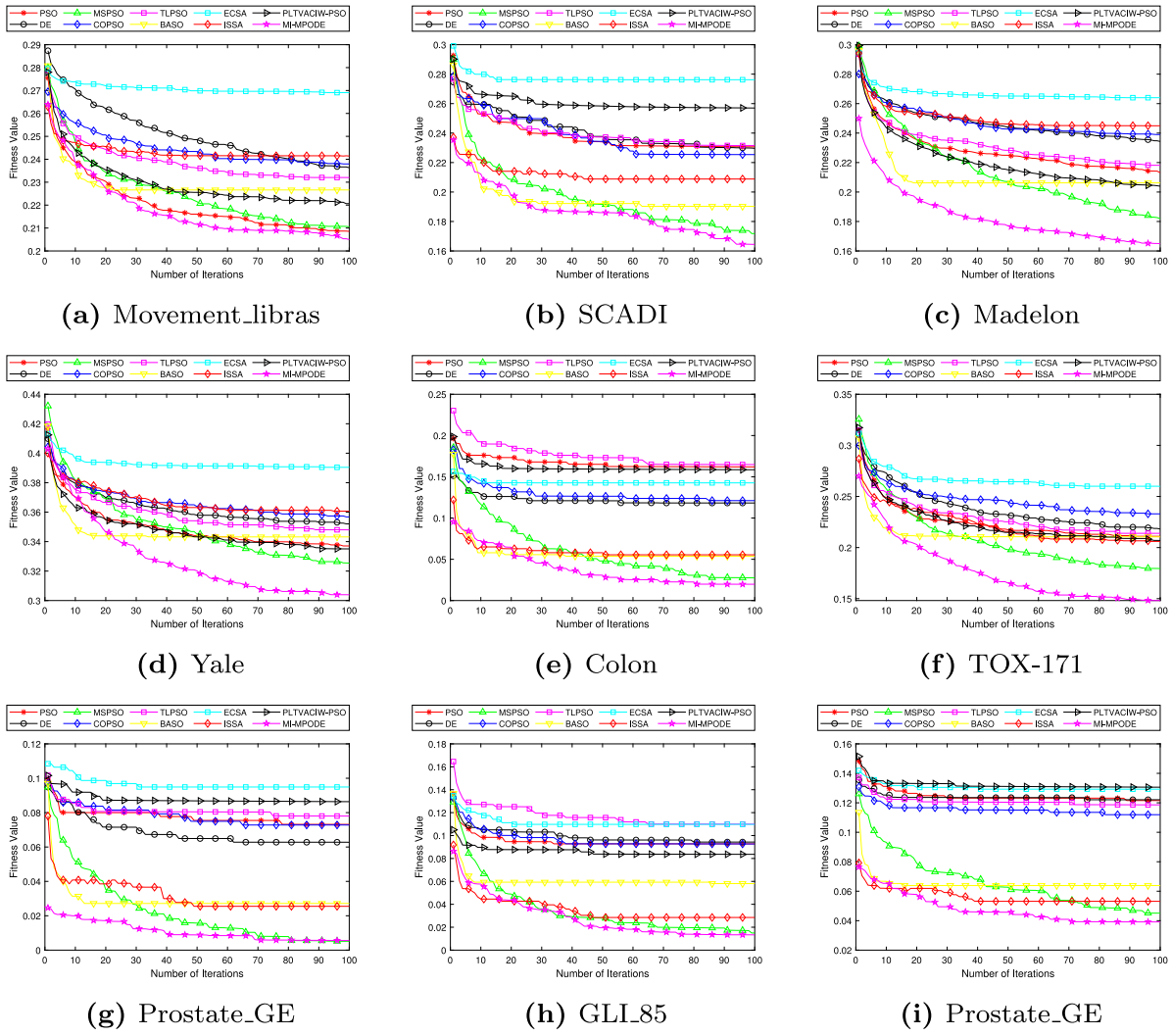


Fig. 5. Convergence curves of 30 independent runs for MI-MPODE and state-of-the-art algorithms on datasets.

Table 7
The wilcoxon-test results for comparing algorithms on datasets.

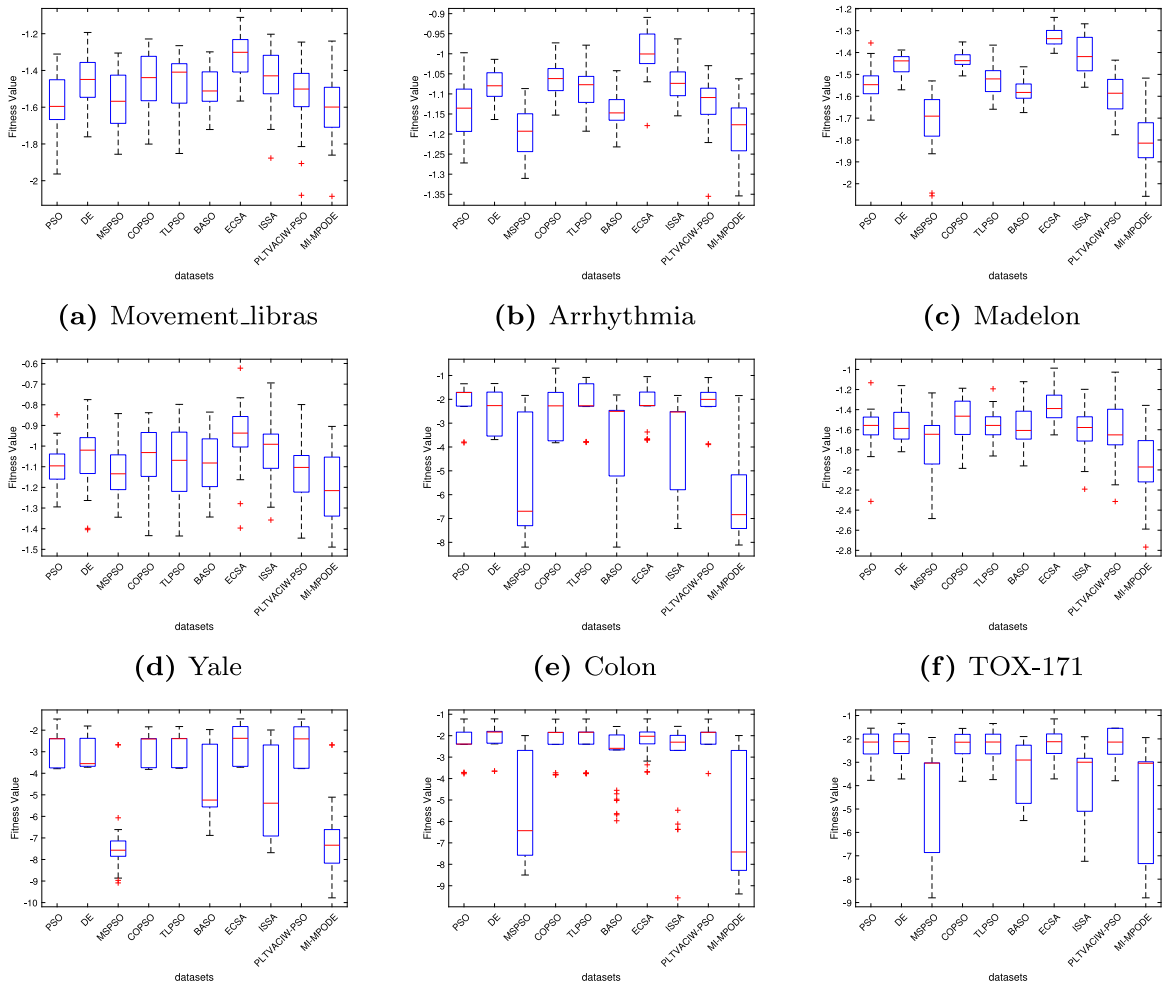
MI-MPODE vs.	R^+	R^-	p-value	Sig.
PSO	252	1	**0.000046**	Yes
DE	241	12	**0.000201**	Yes
MSPSO	108	145	0.548097	No
COPSO	237	16	**0.000334**	Yes
TLPSO	249	4	**0.000070**	Yes
BASO	246	7	**0.000105**	Yes
ECSA	253	0	**0.000040**	Yes
ISSA	250	3	**0.000061**	Yes
PLTVACIW_PSO	250	3	**0.000061**	Yes

Table 8
The friedman-test results for comparing algorithms on datasets.

Algorithm	*AVR*	*Rank*
PSO	7.00	7
DE	7.45	9
MSPSO	1.77	1
COPSO	6.73	6
TLPSO	7.95	10
BASO	5.23	4
ECSA	10.45	11
ISSA	6.68	5
PLTVACIW_PSO	7.45	8
MI-MPODE	2.77	2

superior performance of MI-MPODE mainly demonstrated on the ASU dataset in Section 4.3, and considering the limitations in the length of the paper, the experiments in this section are only conducted on the ASU datasets and a part of the UCI dataset. The results of classification accuracy and the number of selected features are presented in Tables 9 and 10, respectively. In these tables, DE, MPDE, MPODE, and MI-MPODE represent the original algorithm, the algorithm with multiple population mechanism, the algorithm with both multiple populations and LensOBL, and the proposed algorithm in this paper, respectively.

According to the results in Table 9, MI-MPODE demonstrates superior performance among all algorithms. With the introduction of different methods, a noticeable improvement in classification accuracy across multiple datasets can be observed, such as "Arrhythmia", "SCADI", "Madelon", "Yale", "TOX-171", and "GLI_85". This also validates the effectiveness of the proposed method in this paper. Combining with the number of features in Table 10, MI-MPODE also obtains the fewest number of features across multiple datasets. However, the selected number of features do not decrease gradually with the introduction of methods. It is noteworthy that the number of features for MPODE show an increase phenomenon, such as in multiple datasets including "Arrhythmia", "LSVT", "SCADI", "Yale", and "Leukemia". Through research, it is found that the main reason is that MPODE with the LensOBL strategy increases the diversity of the population.



(a) Movement_libras (b) Arrhythmia (c) Madelon

(d) Yale (e) Colon (f) TOX-171

(g) Prostate_GE (h) Leukemia (i) Prostate_GE

Fig. 6. Box curves of 30 independent runs for MI-MPODE and state-of-the-art algorithms on datasets.

However, during the same iteration process, MPODE converges slower than MPDE, resulting in more features obtained in the end compared to MPDE. In contrast, regarding the classification accuracy in Table 9, MPODE outperforms MPDE, which sufficiently demonstrates the effectiveness of LensOBL in improving algorithm performance. Additionally, for easier observation of results, the changes in the number of features obtained by algorithms under different methods is also presented in Fig. 7.

In general, MI-MPODE achieves higher classification accuracy with fewer features, which is closely tied to the multiple population mechanism, LensOBL, and MI techniques introduced in this paper. Therefore, this method saves computational costs without compromising the desired results.

5. Conclusion

As an essential component of data preprocessing, FS enhances classification performance by filtering redundant features, thereby conserving computational resources significantly and providing interpretability to the algorithms. Herein, this paper proposed a hybrid FS method that combines both the filter and wrapper approaches. Specifically, the filter uses MI techniques to select a subset of redundant features from a feature space. This operation reduces the initial feature space of the algorithm, which is beneficial for achieving higher classification performance with fewer features. The wrapping process involved an EC-based FS method. In the EC, this paper introduced a multi-population DE algorithm based on LensOBL and proposed a novel multi-population information-sharing mechanism. This mechanism segments a population into multiple subpopulations, with each subpopulation containing three tiers of individuals: elite, common, and inferior. The three tiers of individuals assume different roles in the population, and information sharing occurs primarily among the common individuals. The common individuals can access information from elite individuals in their current subpopulation as well as from common individuals in other subpopulations. Meanwhile, the elite and inferior individuals in the population can guide the population toward better regions and maintain the population diversity, respectively. Furthermore, based on the characteristics of the different individuals, the corresponding individuals are subjected to enhancement strategies and have different updating methods. Additionally, the LensOBL strategy was employed to further enhance the diversity of the population and improve the convergence speed. The combination of these two factors resulted in MI-MPODE, which ultimately selects the optimal feature subset and classification accuracy using the KNN.

To evaluate the performance of MI-MPODE, it was first compared with state-of-the-art algorithms through experiments on multiple FS datasets, and the results were analyzed. The experimental results demonstrated that MI-MPODE performed well in terms of classification accuracy and the number of selected features. In particular, for the high-dimensional datasets, MI-MPODE significantly outperformed the other algorithms. Second, to validate the effectiveness of the new strategies introduced herein, the paper sequentially introduces

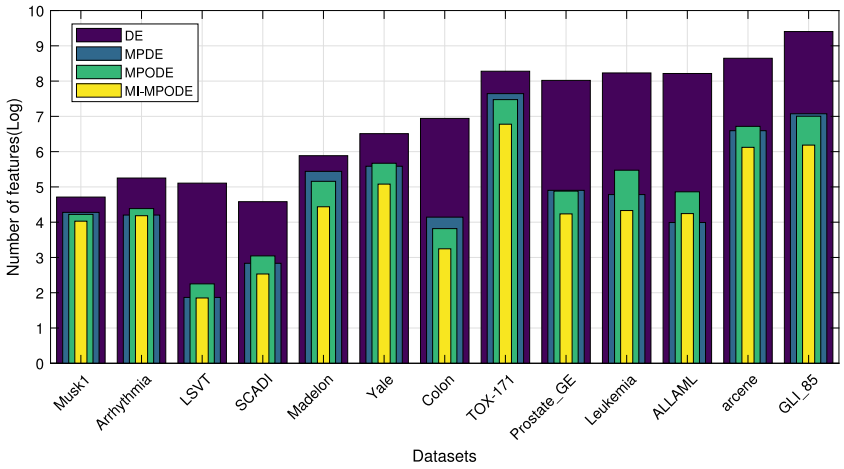


Fig. 7. Bar chart of the number of features obtained by algorithms with different methods.

Table 9
Classification accuracy of algorithms with different methods on the dataset.

Name	Type	DE	MPDE	MPODE	MI-MPODE
Musk1	Mean	0.9366	0.9263	**0.9373**	0.9305
	Std	0.0173	0.0253	**0.0271**	0.0200
Arrhythmia	Mean	0.6778	0.6874	0.6889	**0.6938**
	Std	0.0173	0.0220	0.0189	**0.0208**
LSVT	Mean	0.6784	0.8162	**0.8297**	0.8027
	Std	0.0533	0.0652	**0.0455**	0.0508
SCADI	Mean	0.7833	0.8024	0.8119	**0.8333**
	Std	0.0439	0.0406	0.0546	**0.0342**
Madelon	Mean	0.7912	0.8037	0.8142	**0.8442**
	Std	0.0109	0.0187	0.0254	**0.0171**
Yale	Mean	0.6639	0.6707	0.6850	**0.6966**
	Std	0.0551	0.0460	0.0475	**0.0519**
Colon	Mean	0.9028	0.9806	0.9528	**0.9806**
	Std	0.0792	0.0358	0.0566	**0.0420**
TOX-171	Mean	0.8065	0.8294	0.8458	**0.8601**
	Std	0.0436	0.0528	0.0501	**0.0471**
Prostate_GE	Mean	0.8983	0.9400	0.9450	**0.9600**
	Std	0.0676	0.0578	0.0379	**0.0443**
Leukemia	Mean	0.9619	0.9952	0.9762	**0.9952**
	Std	0.0522	0.0181	0.0433	**0.0181**
ALLAML	Mean	0.8714	**0.9929**	0.9452	0.9643
	Std	0.0687	**0.0218**	0.0584	0.0615
arcene	Mean	0.8742	0.9275	**0.9350**	0.9308
	Std	0.0471	0.0396	**0.0443**	0.0352
GLI_85	Mean	0.9294	0.9784	0.9824	**0.9882**
	Std	0.0644	0.0288	0.0274	**0.0239**

Table 10
Number of features obtained by algorithms with different methods in the dataset.

Name	DE	MPDE	MPODE	MI-MPODE
Musk1	111.30	72.13	68.13	**56.20**
Arrhythmia	190.93	66.83	80.27	**65.60**
LSVT	165.20	6.47	9.50	**6.37**
SCADI	97.63	17.03	21.00	**12.57**
Madelon	359.57	230.67	173.93	**84.53**
Yale	671.80	266.67	290.27	**160.70**
Colon	1035.23	63.00	45.50	**5.67**
TOX-171	3943.40	2090.07	1763.23	**878.23**
Prostate_GE	3041.97	134.73	131.77	**69.07**
Leukemia	3756.17	119.40	238.30	**75.97**
ALLAML	3698.50	**54.07**	129.17	69.67
arcene	5693.80	730.60	826.10	**455.97**
GLI_85	12146.27	1181.03	1101.17	**485.53**

different methods and evaluates their impact on the final results. The results demonstrate the effectiveness of the proposed methods, showing that MI-MPODE can achieve higher classification accuracy with fewer features. However, MI-MPODE still has some limitations. For instance, firstly, MI-MPODE's performance is not significant on small-scale datasets. Secondly, for the two objectives to be optimized in the feature selection problem, namely classification accuracy and the number of features, it is more appropriate to address them as a multi-objective optimization problem.

Therefore, in future work, the algorithm presented in this paper will undergo further optimization and will be used as a multi-objective optimization algorithm to optimize FS problems. Secondly, this paper plans to further investigate the application of multi-population algorithms in FS, particularly for high-dimensional datasets such as text, image, and biological features. Finally, this paper will apply the algorithm to real-world scenarios, such as remote-sensing image classification, text classification, and security detection.

CRediT authorship contribution statement

Fei Yu: Conceptualization, Methodology, Data curation, Supervision, Writing – original draft, Writing – review & editing. **Jian Guan:** Conceptualization, Methodology, Data curation, Software, Writing – original draft, Writing – review & editing. **Hongrun Wu:** Supervision, Writing – review & editing. **Hui Wang:** Supervision, Writing – review & editing. **Biyang Ma:** Supervision, Writing – review & editing.

Declaration of competing interest

No conflict of interest exists in the submission of this paper, and it is approved by all authors for publication.

Data availability

Data will be made available on request.

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